

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3430UA0-1



SCIENCE (Double Award)

Unit 1: BIOLOGY 1
HIGHER TIER

WEDNESDAY, 14 JUNE 2017 – MORNING
1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	10	
3.	6	
4.	8	
5.	8	
6.	10	
7.	7	
8.	6	
Total	60	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 8 is a quality of extended response (QER) question where your writing skills will be assessed.



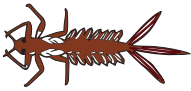
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Answer all questions.

1.

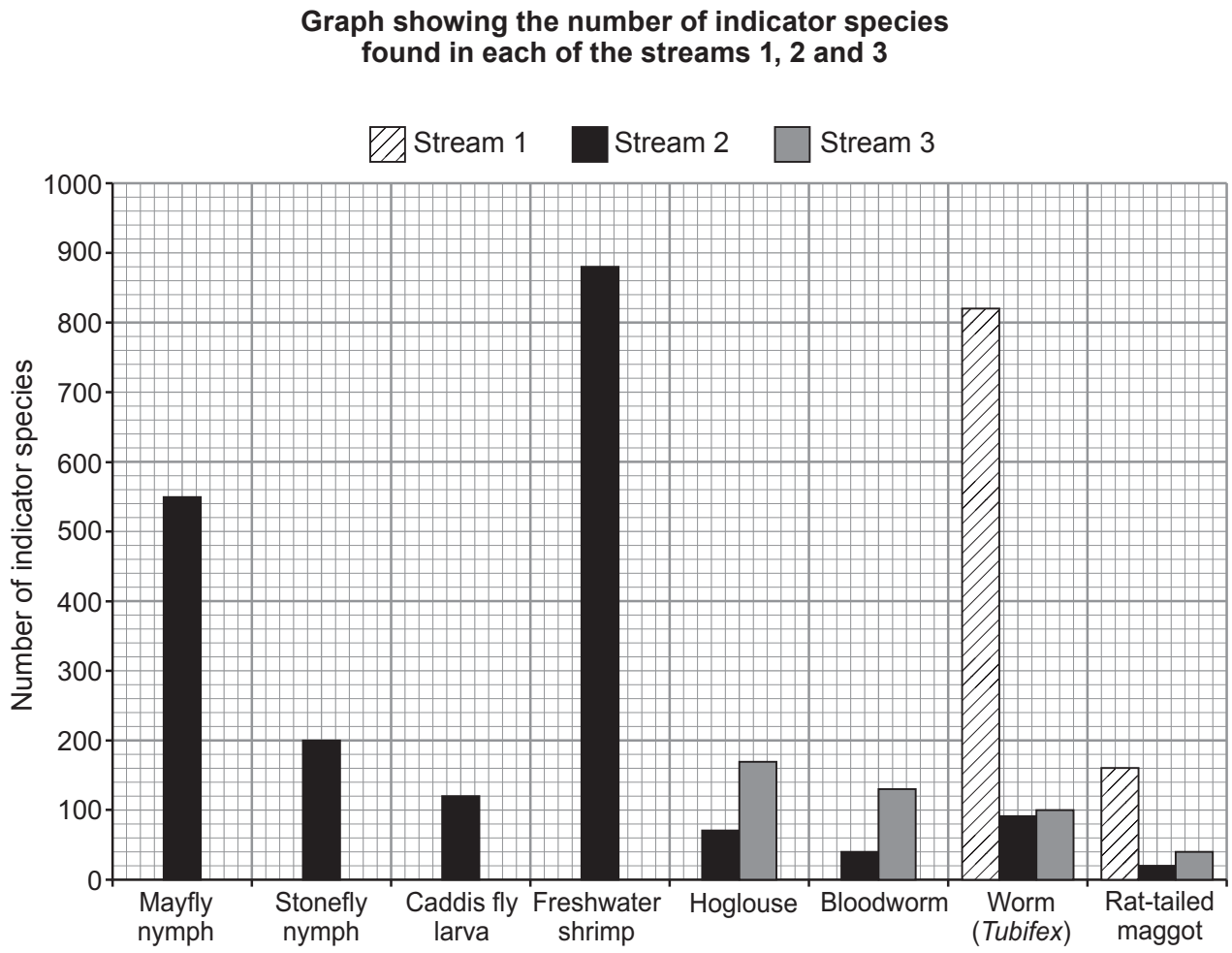


Following flooding in parts of Wales in 2015, environmental scientists investigated nitrate pollution in three different streams. They counted the numbers of eight different indicator species in equal lengths of the three streams, on the same day. The indicator species they counted are shown in the chart below.

GROUP	INDICATOR SPECIES		POLLUTION LEVEL
A	 Mayfly nymph	 Stonefly nymph	If you find these animals in your stream, then there is LITTLE OR NO POLLUTION
B	 Caddis fly larva	 Freshwater shrimp	If you find these animals, but none from Group A , then there may be SLIGHT POLLUTION
C	 Hoglouse	 Bloodworm	If you find these animals, but none from Groups A or B , then there is probably MEDIUM POLLUTION
D	 Worm (<i>Tubifex</i>)	 Rat-tailed maggot	If you find these animals, but none from Groups A , B or C , then there is HIGH POLLUTION
E	No live animals found		If you find no animals at all, then there is VERY HIGH POLLUTION



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(a) (i) Using the graph above and the chart on the opposite page state the pollution level in each of the three streams. [2]

stream 1
stream 2
stream 3

(ii) Suggest **two** possible **sources** of nitrate pollution in streams. [2]

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(b) Name **one** type of indicator species which could be used to assess levels of air pollution. [1]

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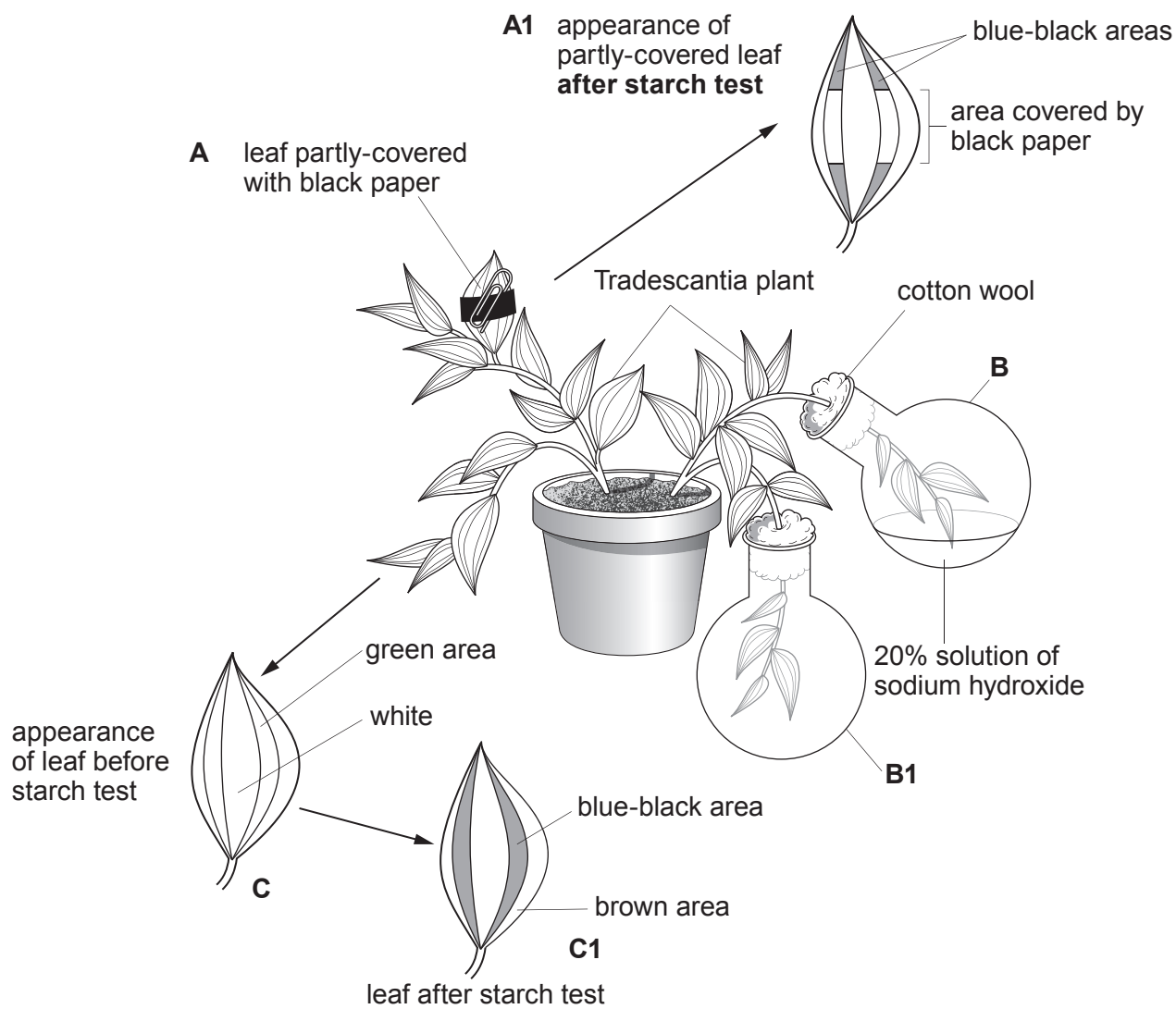


2. *Tradescantia* is a plant whose leaves have green and white areas.



Tradescantia leaves

In the experiment below, a *Tradescantia* plant is used in an investigation to demonstrate **three** factors, **A**, **B** and **C**, needed for photosynthesis.



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(a) (i) State the factor needed for photosynthesis, which is being demonstrated in each of the following: [3]

A

B

C

(ii) How could the experiment in flasks **B** and **B1** be improved? Explain your answer. [2]

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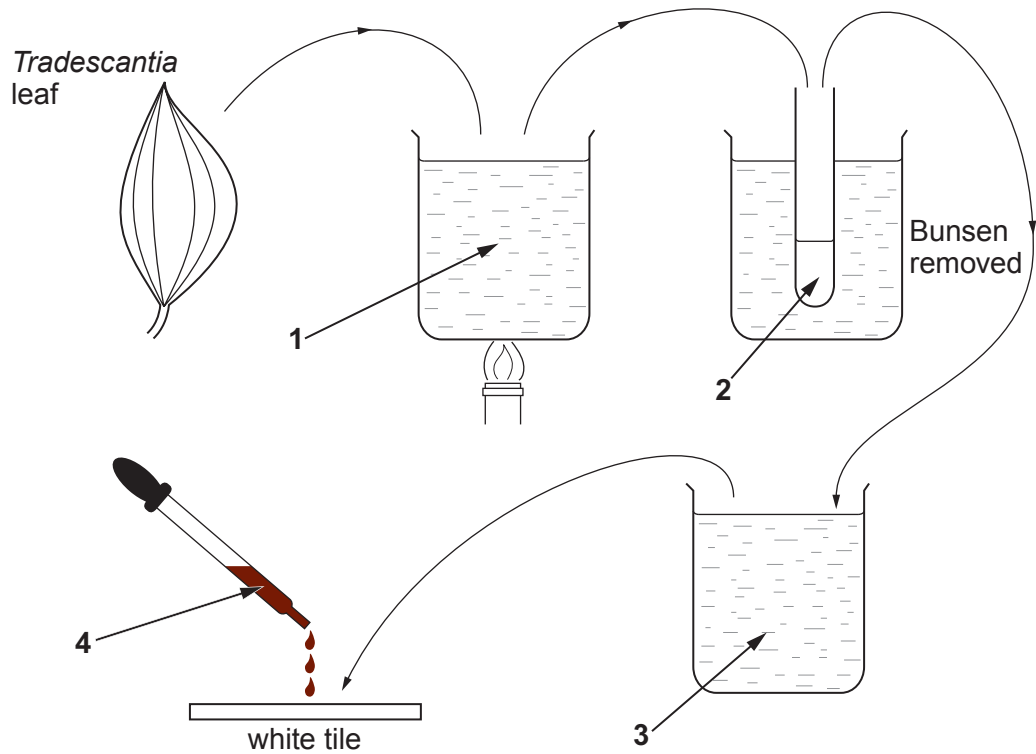
(iii) Before the apparatus was set up, the *Tradescantia* plant was kept in a dark cupboard for 48 hours. Explain the reason for this. [2]

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(b) The diagram below shows the stages in testing a leaf for starch.



Name the liquids **1**, **2**, **3** and **4** shown in the diagram.

[3]

- 1
- 2
- 3
- 4

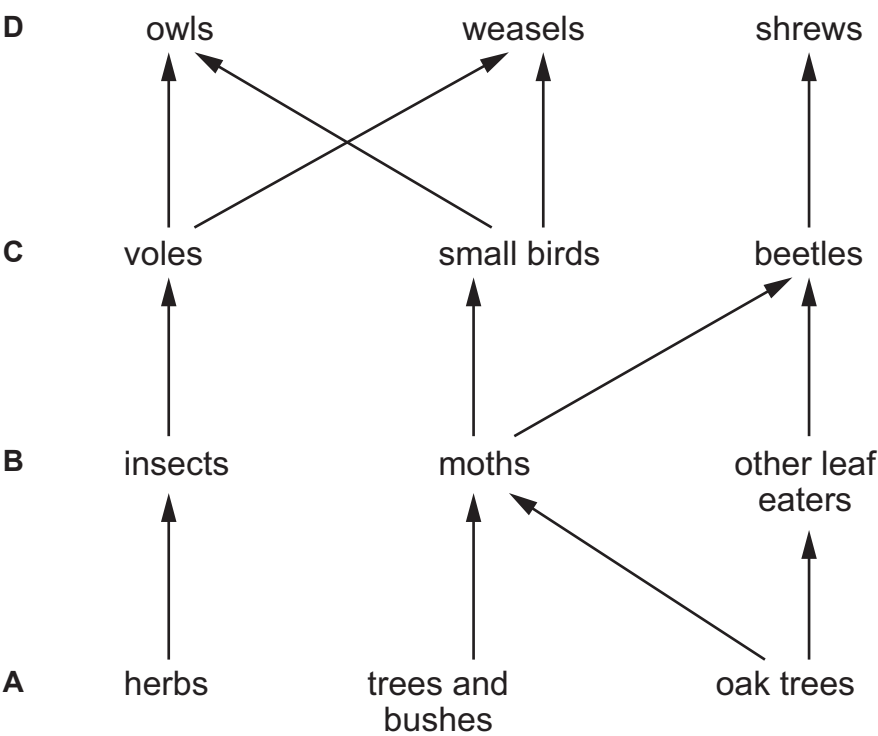


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3. The diagram below shows a food web for a woodland.

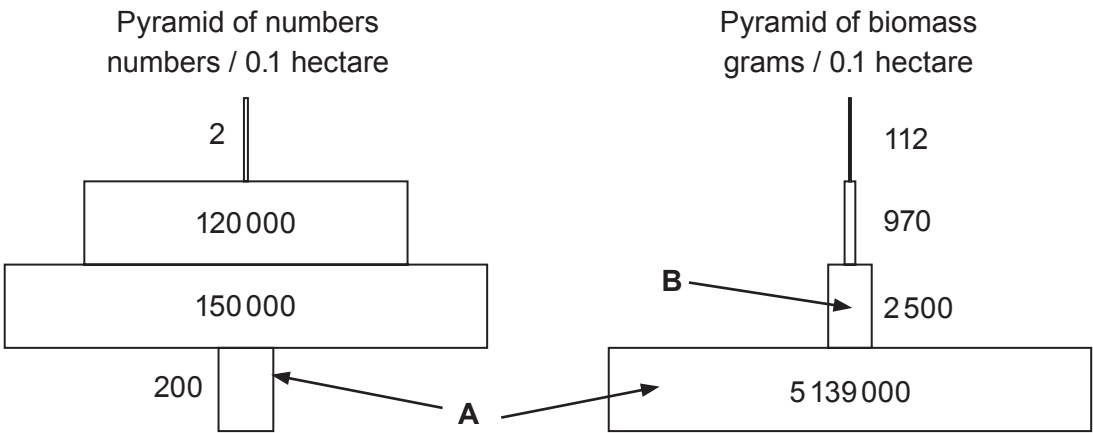


(a) The food web shows four different trophic levels. Give the term used to describe each of the trophic levels. [1]

- A
- B
- C
- D



(b) The diagrams below show a pyramid of numbers and a pyramid of biomass for a food chain in this food web.



(i) Calculate the percentage of biomass passed from trophic level **A** to trophic level **B**.
[2]

Percentage of biomass = %

(ii) Explain why the trophic level labelled **A** is such a different width in the two pyramids.
[2]

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.....

(iii) State **one** way in which energy is lost between trophic levels.
[1]

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4. (a) State the meaning of the term diffusion. [1]

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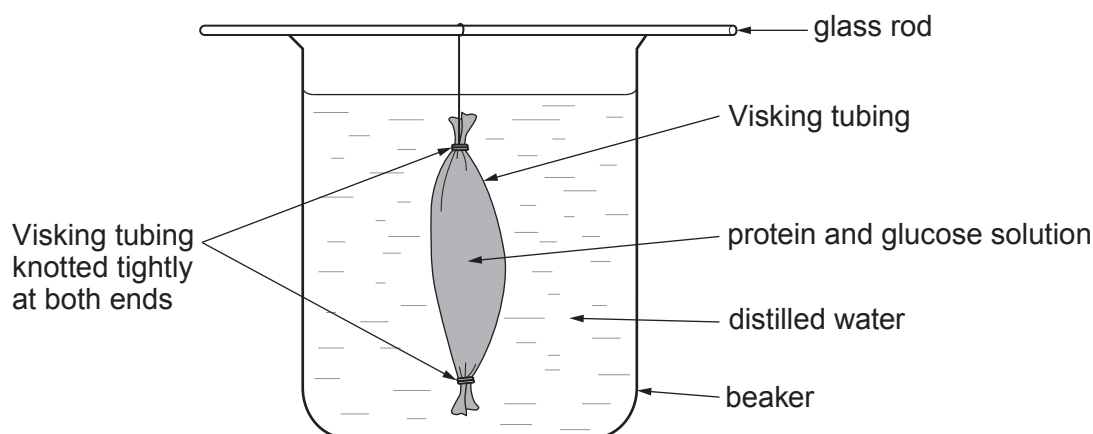
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(b) After a lesson on the properties of cell membranes a year 10 class was asked to investigate some of these properties using Visking tubing. They were given the following instructions:

- Soak a 15 cm length of Visking tubing in water to soften it.
- Tie a knot in one end of the tube.
- Fill the tube with a solution made up of protein and glucose dissolved in water.
- Tie a knot in the open end of the tube.
- Wash the tube under a stream of tap water for 15 seconds.
- Using a glass rod suspend the Visking tubing in a beaker of distilled water.



The diagram below shows how your apparatus should appear.



(i) Why were the students instructed to 'wash the tube under a stream of tap water for 15 seconds'? [1]

.....

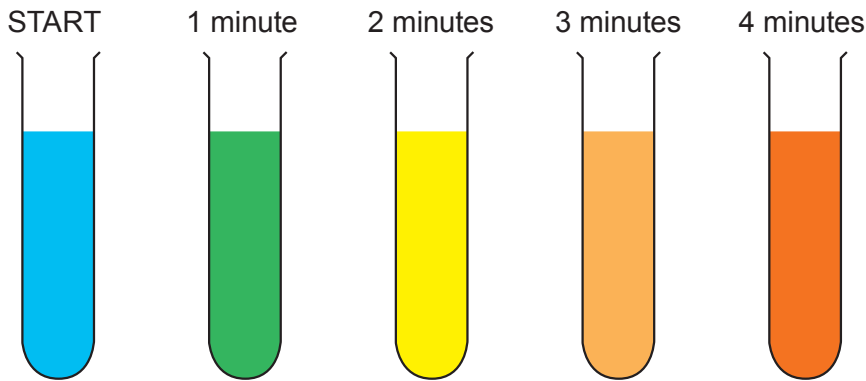
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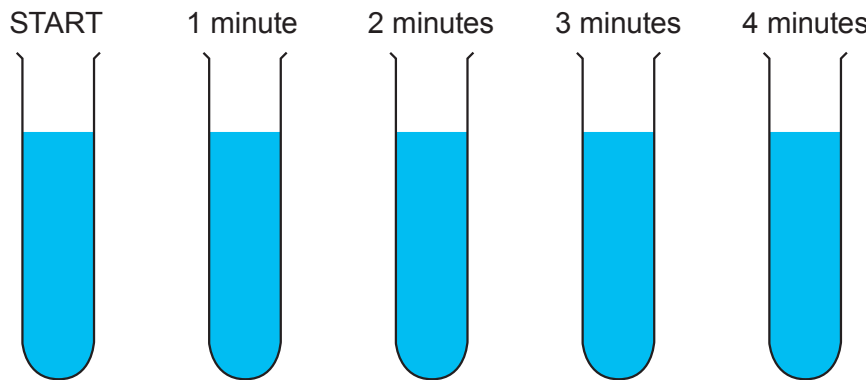
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The students were asked to sample the distilled water **in the beaker** for the presence of both glucose and protein at the start of the experiment and every minute for the next four minutes. Dafydd decided to photograph his results on his smart phone. The photographs he obtained are shown below.

Benedict’s test for glucose



Biuret test for protein



(ii) Explain Dafydd’s results, **at four minutes**, for the contents of the **beaker** for both the Benedict’s test and the Biuret test.

Benedict’s test [3]

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Biuret test [3]

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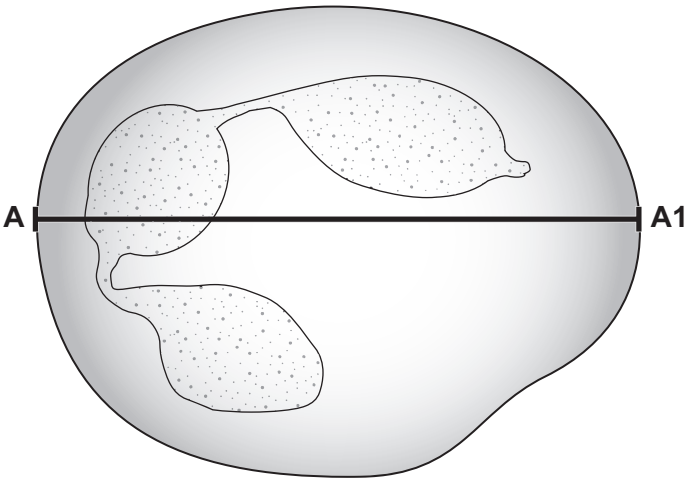


5. (a) Sarah was viewing a smear of blood under a light microscope. How could she calculate the magnification of the microscope she was using? [1]

.....

.....

(b) Sarah produced the drawing below. It shows one of the blood cells she looked at under the microscope.



(i) Name the type of white blood cell shown in the drawing above. [1]

.....

(ii) The actual size of the white blood cell, across the line **A – A1**, was 20 μm (20 micrometres). There are 1000 μm in 1 mm.

I. Measure the length of the line **A – A1** in the diagram above. [1]

length of the line **A – A1** = mm

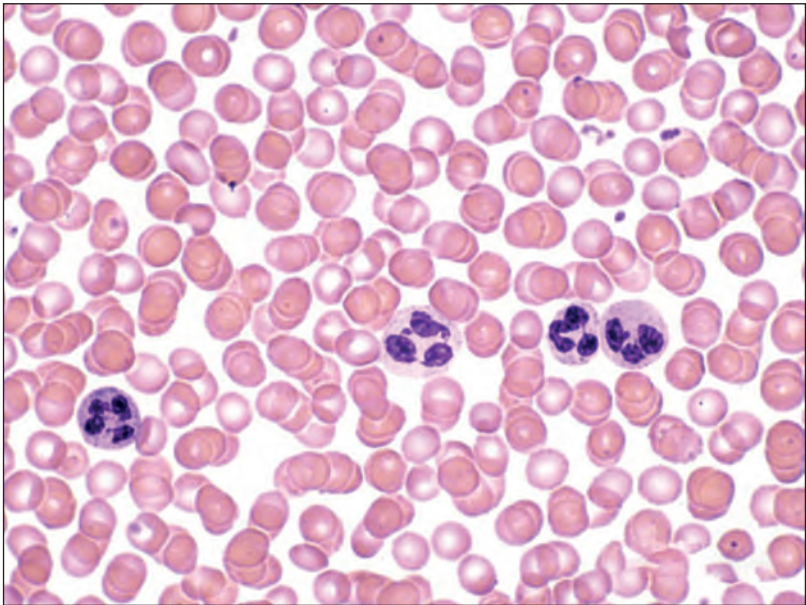
II. Calculate how many times bigger Sarah’s drawing is than the actual blood cell. [2]

Magnification = $\times$



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- (c) The photograph below shows a smear of blood viewed under the high power magnification of a microscope.



- (i) Describe the procedure involved in preparing these blood cells for viewing under the microscope and state the reason why this is needed. [2]

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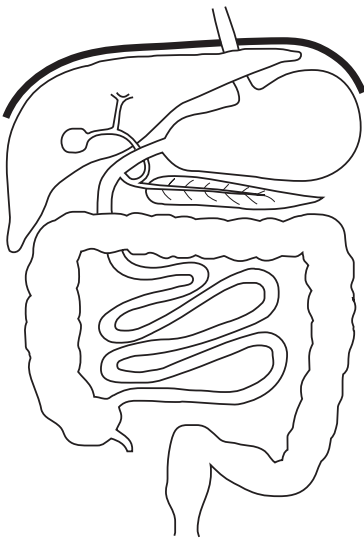
- (ii) The ratio of red blood cells to white blood cells is 500:1. A sample of blood has 57 000 red blood cells. Calculate the number of white blood cells in the sample. [1]

Number of white blood cells =

8



6. The diagram below shows part of the human digestive system.

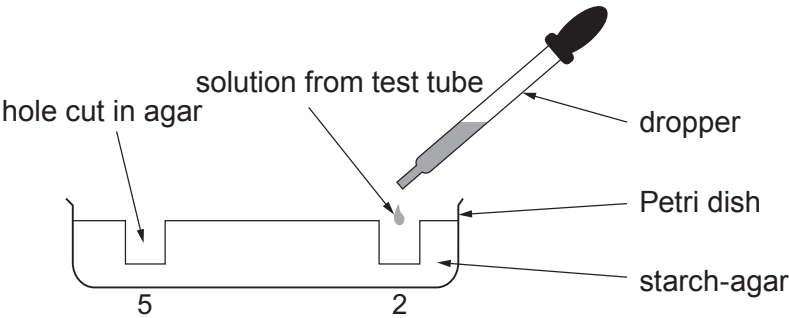


- (a) On the diagram above label the following: [2]
- A** pancreas
B bile duct
- (b) The diagram below shows five test tubes **A – E** and their contents. The pH and temperature of each is also shown.

	Solution				
	Distilled water	0.5% amylase	1% amylase	1% amylase	2% amylase
tube					
temperature (°C)	37	37	100	37	37
pH	7	7	7	7	7

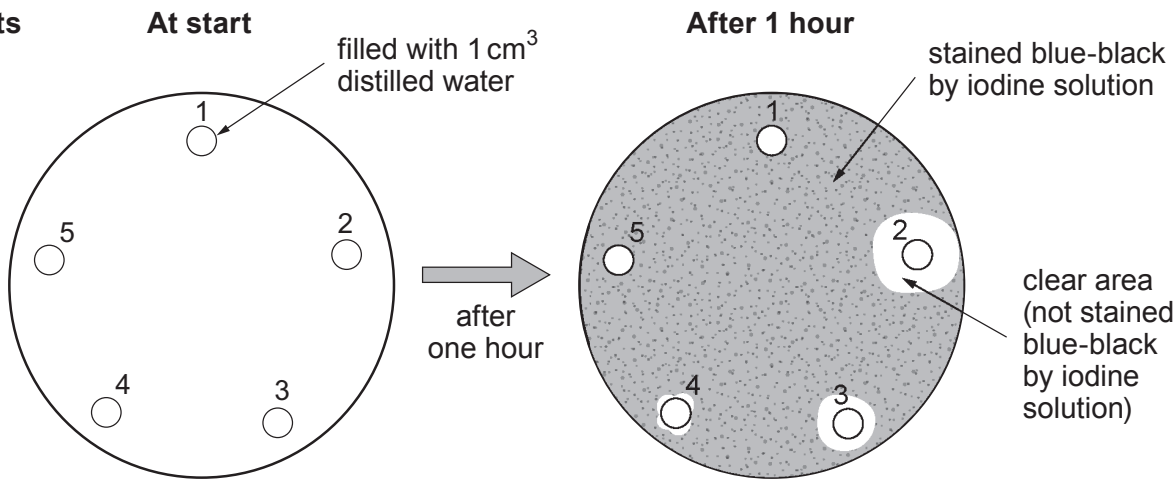
- A pupil carried out the following procedure:
- She took a Petri dish containing starch-agar (a set jelly in which starch had been dissolved).
 - She cut five small holes in the jelly (holes 1 – 5).
 - She put 1 cm³ distilled water from Tube **A** into hole 1.
 - Holes 2 – 5 were each filled with 1 cm³ of a different solution from one of test tubes **B**, **C**, **D** or **E**.
 - After one hour she flooded the Petri dish with iodine solution.





Cross-sectional diagram showing two of the five holes.

Results



(i) Explain the appearance of the clear areas. [1]

.....

.....

(ii) State the solution that was added to hole 2. Explain your answer. [2]

.....

.....

(iii) Which hole contained the contents of tube C? Explain your answer. [4]

Hole number

Explanation

.....

.....

.....

(c) State the purpose of tube A, the contents of which was put into hole 1. [1]

.....

10



7. The heart is a pump which supplies blood to the organs of the body. It deals with increasing physical activity, such as exercise, by increasing its rate (number of beats per minute). After exercise, the heart rate will eventually return to the resting rate. The time the heart takes to return to its resting rate is a good indication of a person's level of fitness.

A researcher wanted to investigate the difference in the level of fitness between men and women. Six people at a gym volunteered to compare their fitness. Each volunteer ran on a treadmill for five minutes. Immediately at the end of the exercise their heart rates were recorded on a laptop computer. Their heart rates were also recorded one minute later.



The researcher conducting the investigation used the data collected from each volunteer to calculate their fitness index. An index of 60 – 70 shows a person who is reasonably fit, below this level shows decreasing fitness.

The results of the investigation are shown in the table below.

Volunteer	Resting heart rate (beats/min)	Heart rate immediately after exercise (beats/min)	Heart rate 1 min after exercise (beats/min)	Fitness index	Body mass (kg)
John	70	190	130	35	56.5
Michael	70	150	80	60	63.6
Charlotte	60	110	65	75	76.4
Paul	80	195	140	20	101.8
Peter	75	180	156	43	59.1
Lucy	70	130	80	68	56.1

(a) (i) Who appears to be the least fit person in the table? Explain your answer. [1]

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(ii) What is the relationship between **fitness index** and heart rate immediately after exercise? [1]

(iii) At the end of the investigation the volunteers discussed the results. Peter said that “he thought that John was fitter than Lucy because one minute after exercise John’s heart rate had fallen by 60 beats/min whereas Lucy’s had only fallen by 50 beats/min.” The researcher said that you must calculate the percentage decrease in the heart rates in order to make a fair comparison.

I. In the space below calculate the percentage decrease in the heart rates of both John and Lucy. [2]

John

Percentage decrease in heart rate =

Lucy

Percentage decrease in heart rate =

II. From the answers to (I.) above state whether John or Lucy appears fittest. [1]

.....

(b) The design of this investigation has many faults.

Give **two** factors which should be controlled in this investigation. [1]

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(c) Why is it not possible to conclude from the data that Paul’s low **fitness index** is due to obesity? [1]

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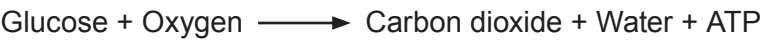
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8. The following word equations show the two types of cell respiration which occur in humans.

Equation 1



Equation 2



Name each of the types of cell respiration shown above and write an account explaining when each occurs in the human body. Include any advantages or disadvantages of each type of respiration. [6 QER]

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